Physics-based vs surrogate contact pressure models for musculoskeletal simulation

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Introduction

Predictive musculoskeletal simulations can be a valuable tool for estimating how rehabilitation treatments or surgical interventions influence an individual's optimal movement, particularly in individuals with osteoarthritis. These simulations typically require around 30 min for a full-body model on a conventional laptop [1]. Recently, our group developed a method to incorporate knee contact pressure models into these simulations, enabling the optimization to minimize contact pressure in specific regions of the joint [2,3]. The resulting optimal control problems, solved using automatic differentiation, remained computationally tractable, requiring 1–3 h on a standard laptop, compared with approximately 30 min without contact pressure models. However, further reductions in computational time are needed to bring these simulations closer to clinical application. To this end, we replaced the smooth collision detection algorithm with a neural network–based surrogate model and compared the computational performance of the two approaches.

Methods

Different tracking simulations were formulated to evaluate the effect of using a surrogate model in musculoskeletal simulations. Experimental data from the Knee Grand Challenge were used as reference [4]. These data included medial and lateral knee contact forces measured in a subject with a knee prosthesis. Joint coordinates, joint velocities, ground reaction forces, and knee contact forces were tracked while estimating muscle forces and knee contact pressures, allowing slight deviations from the experimental data to ensure dynamic consistency.

The simulations differed only in the computation of knee contact pressures. In optimization A, pressures were computed using our developed smooth collision detection algorithm [3], which evaluates the pressure at each mesh cell of the contacting geometries (tibial tray and femoral component) based on the joint kinematics and continuously differentiable penetration functions.

In optimization B, knee contact pressures were estimated using a feedforward artificial neural network (ANN) surrogate model trained on data generated by the physics-based model. The trained ANN was then converted into a CasADi function and used to estimate joint pressures at each optimization iteration.

Results

The results show that the optimal control problem achieved a lower cost function value when using the surrogate contact model, although it required a longer computational time to converge. Tracking of knee contact forces was slightly more accurate with the physics-based model, whereas tracking of the GRFs was slightly more accurate with the ANN (Figure 1, Table 1).

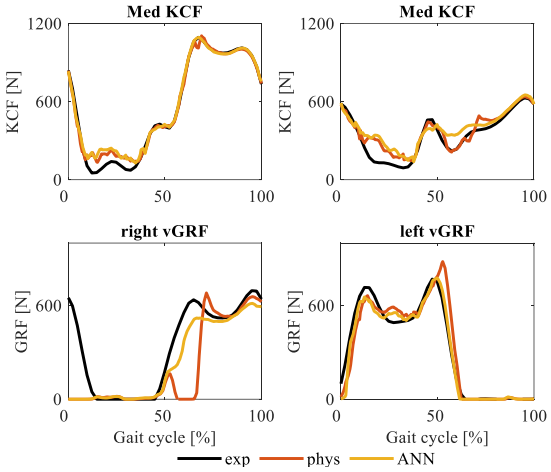


Figure 1: Tracking of the medial (Med) and lateral (Lat) knee contact forces (KCF) and GRF for the physics-based (phys) and the ANN models.

Model		A: Physics model	B: ANN
Cost function value		1.43	1.19
Computational time		1h 11 min	4 h 41 min
RMSE	Med KCF [N]	48.1	55.5
	Lat KCF [N]	63.0	81.7
	GRF [N]	72 ± 92.8	50.0 ± 58.8
	Qs [º]	1.3 ± 2.2	1.1 ± 1.8

Table 1: Main results of the tracking simulations comparing the performance using the physics-based model and its surrogate model. Med and Lat KCF stand for medial and lateral knee contact forces.

Conclusions

This study demonstrates the feasibility of using a surrogate contact model within a tracking simulation. Future work will focus on reducing the computational time.

Acknowledgements

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